Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment: Affective Constructs Are a Bottleneck, Not a Basis, for Depression

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Abstract

We study whether graph-guided mixture-of-experts (MoE) routing can improve multimodal mental-health assessment. We evaluate a unified model on DAIC-WOZ depression detection, CMU-MOSEI sentiment/emotion, and ChaLearn First Impressions personality, combining modality-specific encoders, gated late fusion, an MMoEEx expert bank, and a KNN-graph GraphSAGE router. Across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned per-task routing weight, routing does not improve depression detection. A homophily diagnostic localizes why: the graph actually used for routing, built on an untrained projection of raw features, not the model's trained representation, carries no depression-label signal beyond chance, though the trained representation itself does for inductive constructions; sentiment and personality carry real signal under either. Separately, we examine how the trained representation itself encodes depression: three 12-dimensional projections under an identical protocol, a supervised affective-construct projection, a random projection, and PCA, with no graph or router involved. Random projection and PCA are indistinguishable from each other and both exceed the construct projection, never separable from a label-permutation null; the gap is negative in every seed under a subject-level bootstrap. Affective-construct projections achievable at clinical sample sizes, transferred across corpora or estimated in-domain, degrade depression-relevant variance an unsupervised projection retains. An independent corpus dissociates the cause: ground-truth Big-Five traits decode depression; traits estimated from the same features do not generalize. The bottleneck is construct measurement, not construct validity. These two findings share a protocol and dataset, not a causal chain, and point to a common theme: supervised structure fit at small clinical sample sizes can destroy more than it organizes. We release the leakage-safe protocol and the ablation ladder.

1 Introduction

Multimodal mental health assessment integrates verbal and non-verbal signals — language, prosody, facial expressions — to predict clinically relevant constructs such as depression severity, emotional state, and personality traits. Prior work tackles these tasks in isolation with task-specific architectures that cannot share complementary information across related supervision signals. Multitask and multimodal fusion approaches exist, but often yield opaque models whose routing decisions are difficult to interpret, and rarely address label leakage introduced by graph-based sample relationships.

This paper asks whether graph-guided MoE routing improves multimodal mental-health assessment when graph construction is made leakage-safe and evaluation is performed under strict subject-independent splits. We evaluate a unified architecture combining modality-specific encoders, a gated late-fusion mechanism, an MMoEEx expert bank, and a graph-guided router built on a leakage-safe K-nearest-neighbour similarity graph over multimodal embeddings. Across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned per-task routing weight, this routing does not improve depression detection over a non-graph MMoEEx baseline. A graph-homophily diagnostic localizes why: the graph actually used for routing is built on a fixed, untrained projection of raw features rather than the model's own trained representation, and this untrained graph carries no depression-label signal beyond chance for DAIC, while carrying real signal for MOSEI sentiment/emotion and FI personality. Recomputing the diagnostic on the trained representation instead shows a more specific pattern: DAIC signal is present for inductive construction and absent for split-local, so "the embedding space never separates depression" is not the full story either — the routing failure and the representation's own structure are separate questions, which we return to directly.

This paper's second, independent question concerns that trained representation on its own terms, with no graph or router in the loop: does supervising it with affective constructs help or hurt depression detection? Under an identical protocol, we compare three 12-dimensional projections of the model's fused DAIC representation: a supervised affective-construct projection (the per-task heads' own outputs), an unsupervised random projection, and unsupervised PCA. Random and PCA are statistically indistinguishable from each other, and both exceed the construct projection, which never separates from a label-permutation null. The gap is negative in every one of five seeds, survives dropping near-constant profile dimensions, and remains negative under a subject-level bootstrap propagating training-seed and test-subject sampling uncertainty. A separate in-domain re-derivation, intended to test whether cross-corpus transfer explains the gap, is itself inconclusive under proper seed replication (Section 5.1) — so this alternative is not ruled out by that control alone. What the evidence supports is narrower than construct supervision failing in principle: affective-construct projections achievable at clinical sample sizes, in this setting, degrade depression-relevant variance that an unsupervised projection of the same features retains. This finding and the routing result above share a protocol and a dataset, not a causal chain — Section 6 discusses why we read them as two instances of a common theme rather than one explaining the other.

This raises a follow-up question: are affective constructs simply uninformative about depression, or is the problem specifically that our estimates of them are unreliable at clinical sample sizes? A second, independent corpus (MPDD) dissociates the two. Ground-truth Big-Five personality scores decode depression well above chance; the same traits, estimated from that corpus's own audio/video features via a regressor trained the ordinary way, fail to generalize to held-out subjects at all (held-out R^2 negative for all five traits). The construct is informative; our estimate of it is not. We read the DAIC result through this lens: the bottleneck is construct measurement, not construct validity — a claim consistent with, rather than contradicting, the clinical literature linking personality and affect to depression (Kotov et al. 2010; Clark and Watson 1991).

Our contributions are threefold. First, a leakage-safe experimental protocol for graph-based routing in multimodal clinical settings, together with a rigorous negative result and a homophily diagnostic that localizes it to the untrained embedding actually used for routing, distinct from the trained representation's own structure. Second, a matched-dimensionality comparison showing that affective-construct projections achievable at clinical sample sizes destroy depression-relevant variance relative to unsupervised projections of the same features, replicated across five seeds with the uncertainty of both training and test-subject sampling accounted for. Third, an independent-corpus dissociation localizing this failure to construct estimation rather than construct validity.

Figure 1 details the software architecture of our experimental pipeline, moving from data extraction through unimodal baselines (Phases 1–4) into the joint multitask training implementation (Phase 7). To isolate the routing effect, we compare five graph-routing ablation variants (V0–V4, evaluating inductive, split-local, and transductive constructions) against non-graph MMoEEx and KNN-voting baselines. None of the routing deltas on DAIC or MOSEI are reportable at n=5 seeds (Section 5); FI personality is the one directional effect, and is reportably degraded. The homophily diagnostic above, a K-sensitivity sweep ($K = 5, 10, 15, 20$), and a learned per-task λ weight in place of the fixed default of 0.5 together localize and probe this null; neither targeted fix recovers a DAIC benefit, consistent with the diagnosis that the routing graph itself lacks the relevant signal rather than the router failing to exploit signal that is present.

2 Related Work

Multimodal fusion for affective computing spans early fusion (feature concatenation), late fusion (learned per-modality gates), and low-rank or cross-attention fusion. Gated late fusion with learned modality gates (Matsumoto et al. 2026) and Low-Rank Multimodal Fusion (Liu et al. 2018) avoid the parameter cost of full tensor fusion. Cross-modal attention has also been proposed for depression detection (Matsumoto et al. 2026; Hua et al. 2026); a single-seed fusion ablation in development did not observe a benefit from it under our evaluation protocol (supplementary material).

Mixture-of-experts. The sparsely-gated MoE layer (Shazeer et al. 2017) routes inputs through a learned subset of experts. Multitask MoE (MMoE) adds task-specific gates (Ma et al. 2018), and MMoEEx adds expert exclusivity/orthogonality regularization to reduce expert collapse (Aoki, Tung, and Oliveira 2022). We build on MMoEEx and add graph-based routing.

Graph neural networks for routing. GraphSAGE (Hamilton, Ying, and Leskovec 2017) learns inductive aggregator functions that generalize to unseen nodes — a requirement for clinical deployment on new participants. We use GraphSAGE (and, as an ablation, GAT (Velickovic et al. 2018)) as a router: aggregating features from a sample's K nearest neighbours to produce a routing vector that modulates expert weights.

LLM-enhanced encoders. Instruction-tuned LLMs and multi-modal LLMs integrating audio and visual understanding have been applied to depression detection (Zhao et al. 2025). We explored LLM-based encoder replacements during development (single-seed, supplementary material only) but do not report them as a confirmatory result here (Limitations).

Trait and state psychology. Personality traits and depressive states are correlated but conceptually distinct (Clark and Watson 1991; Kotov et al. 2010; Bylsma, Morris, and Rottenberg 2008); RDoC (Insel et al. 2010) makes the point structurally: a construct and one operationalization of it are not the same thing. Our MPDD dissociation (Section 5.2) is an empirical instance of this.

Depression detection on DAIC-WOZ. The corpus (Gratch et al. 2014) contains 189 clinical interviews with PHQ-8 labels. Prior work spans graph-based text models (Burdisso et al. 2023), multimodal behavioral phenotyping (Chen and Huang 2026), and gender-emotion interaction multi-task networks (Xing et al. 2026). Closest to our multi-construct framing, PsyGAT (Vyalla et al. 2026) builds a psychologically-grounded, interpretable graph model that explicitly distinguishes

Figure 1: Implementation details of the experimental pipeline. The diagram shows the progression from dataset caching through the non-graph MMoEEx baseline into the Phase 7 multitask epoch loop (5-seed protocol, K-sensitivity sweep, learned routing weight), feeding the central-result analysis (construct-profile comparison, MPDD dissociation, 5-seed reportability statistics). LLM ablations, domain adaptation, calibration, and XAI (dashed) are single-seed exploratory material.

personality-trait context from acute depression symptoms, though it is not evaluated on our datasets or splits and we do not report a direct numeric comparison. Reported metrics vary widely by evaluation protocol (F1 vs. AUROC, inclusion of therapist/interviewer prompts, differing label thresholds); we report AUROC under our own fixed, leakage-safe protocol throughout.

3 Method

We use standard multimodal encoders and MoE components so that any performance change can be attributed to graph-guided routing rather than to a novel backbone. Figure 2 gives an overview: modality-specific encoders project to a common d=256 space; gated late fusion combines them; an MMoEEx expert bank and a KNN-graph GraphSAGE router jointly determine expert routing weights; task heads predict depression, sentiment, emotion, and personality.

3.1 Modality Encoders and Gated Fusion

Text is encoded with RoBERTa-base (Liu et al. 2019) (768-dim), audio with WavLM-large (Chen et al. 2022) (1536-dim; eGeMAPS (Eyben et al. 2015) for the classical-encoder ablation), and video with ViT-B/16 (Dosovitskiy et al. 2021) (1536-dim; OpenFace (Baltrusaitis et al. 2018) action units as an alternative). Each is linearly projected to d=256. Given projected embeddings $h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}$ and a modality-availability mask m_i, gated late fusion computes:

$$\alpha_i = \mathrm{softmax}\big(W_\alpha[h_i^{\text{text}}; h_i^{\text{audio}}; h_i^{\text{video}}]\big),$$
$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon}[m]\, h_i^m.$$

Critically, m_i is not only a missing-data indicator: each dataset is assigned a fixed native routing — DAIC text-only, FI video-only, MOSEI full multimodal — reflecting which modalities are actually informative for that task in our data, not merely which are physically present. This is a deliberate architectural choice, not an artifact: a single-seed fusion ablation in development (Phase 4, gated late fusion / LMF / cross-attention, supplementary material) reached DAIC AUROC 0.38–0.44 with audio/video unmasked, well below the 0.584 text-only baseline used throughout this paper, so the released model runs DAIC as an effectively unimodal-text task inside a jointly-trained multimodal, multitask architecture. We state this here rather than leaving it to be discovered downstream, since it directly qualifies what "multimodal" means for the depression task and is load-bearing for how the graph-routing results below should be read.

3.2 MMoEEx Expert Bank and Graph-Guided Routing

We instantiate M=8 expert MLPs with residual connections, hidden dimension 256 in both configurations we report. For the graph-routed configuration (our primary reported results), all 8 experts are available to every task via a full per-task softmax gate $g_t(h_i) \in \mathbb{R}^8$; we additionally report a non-graph MMoEEx baseline that hard-isolates experts per task group instead, used to quantify the value added by graph routing.

We construct an undirected KNN graph over fused embeddings using cosine similarity, with three leakage-safe construction modes: inductive (graph on training data only; validation/test nodes connect solely to training neighbours), split-local (separate graphs per split, no cross-split edges), and transductive (graph over all splits — ablation only, explicitly not leakage-safe). The embeddings used for graph construction come from a fixed, separately-initialized gated-fusion projection of the raw modality features rather than the jointly-trained model's own h_i^{fused} (Section 3.4) — a deliberate decoupling of graph topology from the evolving task model, but one whose embedding space is a random projection rather than a pretrained or task-relevant one, which we return to when interpreting the homophily diagnostic (Section 5). A two-layer GraphSAGE router aggregates neighbourhood features into a routing vec-

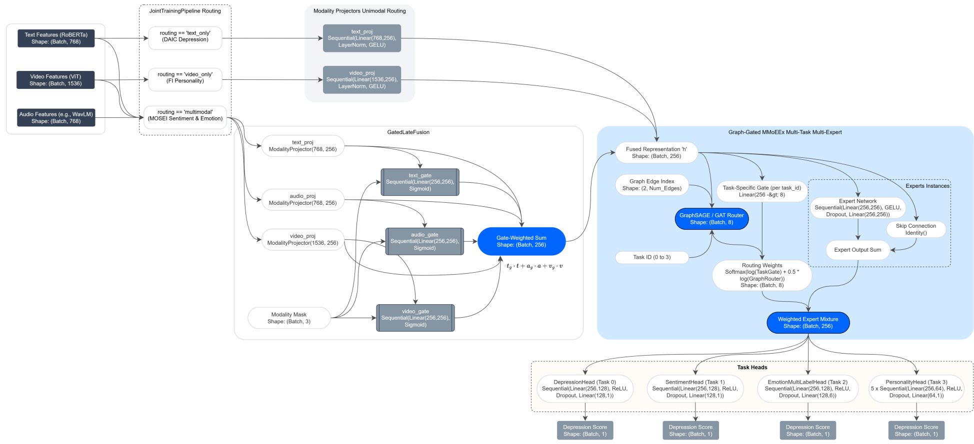

Figure 2: Unified multimodal graph-gated MoE architecture, including per-dataset routing, the gated fusion mechanism, the MMoEEx expert bank, and the GraphSAGE/GAT router.

tor $r_i = \mathrm{softmax}(W_r h_i^{(2)}) \in \mathbb{R}^8$. Task gate and graph router combine in log-space:

$$\tilde{w}_{t,i} = \mathrm{softmax}\big(\log g_t(h_i) + \lambda \log r_i\big),$$

with λ=0.5 fixed across all reported experiments (not tuned).

3.3 Task Heads and Training Objective

Depression (DAIC) uses a binary head with BCE loss; sentiment (MOSEI) a regression head with NLL loss; emotion (MOSEI, 6 labels) a multi-label BCE head; personality (FI, Big-Five) five regression heads with NLL loss. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form (Kendall and Gal 2017) with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \tfrac{1}{2\sigma^2}\big(\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}\big) + \log \sigma.$$

3.4 Construct Profiles and Unsupervised Projections

To test whether h_i^{fused}'s organization around affective constructs is itself the obstacle, we compare three 12-dimensional projections of it, with no graph or router involved: (i) a construct profile — the sentiment (1-dim), emotion (6-dim), and personality (5-dim) heads applied to h_i^{fused} under DAIC's native text-only routing (depression excluded, as the prediction target); (ii) a random Gaussian projection (20 draws); (iii) PCA fit on train. All three feed an identical ℓ_2-regularized logistic regression (nested 5-fold CV) predicting DAIC depression on the held-out test split.

Three controls accompany this: a label-permutation null (1000 permutations) establishes above-chance decoding; a degeneracy control drops near-zero-variance profile dimensions (random projection regenerated at matching dimensionality), ruling out collapsed construct-head outputs; an in-domain control re-derives sentiment/emotion supervision on DAIC transcripts directly (off-the-shelf classifiers, Section 5.1) rather than transferring MOSEI-trained heads, testing cross-corpus domain shift. Both derived-predictor controls require clearing a held-out generalization gate before use; Section 5.1 reports what happens when this check is skipped.

3.5 Independent-Corpus Dissociation (MPDD)

MPDD (Fu et al. 2025) provides actual Big-Five scores and binary depression labels for two age tracks (Young and Elderly; using audio and video features, with no text). Because MPDD's encoders differ from DAIC's, we do not transfer the model directly. Instead, we test the core question directly on MPDD: comparing depression decoding using ground-truth Big-Five scores versus scores estimated by a ridge regressor on MPDD's own features. Evaluation is at the subject level, since segment-level scoring would artificially duplicate each subject. When combining both age tracks, we report a grouped AUC (comparing pairs only within the same track) because a simple pooled ranking would allow age differences to artificially inflate the results.

4 Experimental Setup

4.1 Datasets

DAIC-WOZ: 189 clinical interview sessions (107 train / 35 val / 47 test), PHQ-8 binary depression label, subject-independent splits. CMU-MOSEI (Zadeh et al. 2018): 22,777 utterance-level samples (16,265 / 1,869 / 4,643), continuous sentiment and 6-label emotion. ChaLearn First Impressions (Escalante et al. 2020): 10,000 clips (6,000 / 2,000 / 2,000), Big-Five apparent-personality regression; personality is a distinct construct from clinical depression and serves as auxiliary supervision only.

4.2 Training Details

AdamW (lr=3×10^{-4}), cosine annealing, early stopping (patience 20 epochs, max 150). Batch size 32. Temperature-balanced sampling (T=3.0) prevents MOSEI (120× larger than DAIC) from dominating joint training. Projectors are frozen for the first 20 epochs, then the top 2 layers are unfrozen. All experiments, including LLM-based ablations, were trained on a single NVIDIA RTX A6000 GPU.

4.3 Evaluation

DAIC: AUROC (primary) and F1. MOSEI sentiment: CCC (primary). MOSEI emotion: macro-AUROC over 6 emotions. FI: mean CCC over five traits. All results report 95% BCa bootstrap confidence intervals; AUROC comparisons use the DeLong test; CCC/F1 comparisons use paired permutation tests; effect sizes are Cohen's d.

5 Results

5.1 Construct-Aligned Projections Underperform Unsupervised Ones

Across 5 independently trained seeds of the non-graph MMoEEx baseline, the 12-dimensional construct profile (Section 3.4) decodes DAIC depression at mean test AUROC 0.524 ± 0.039, never separating from a label-permutation null (per-seed permutation $p \in [0.30, 0.75]$) — an absolute claim this design has power only to bound loosely at n_{test}=47. The comparison that is well-powered is relative: a random 12-dimensional projection of the identical fused representation reaches 0.616 ± 0.017, and PCA-12 (unsupervised, variance-preserving, fit on train) reaches 0.610 ± 0.040 — indistinguishable from each other, and both above the construct profile in every seed. Table 1 reports the matched-dimensionality comparison after dropping per-seed near-constant profile dimensions (a variance-based check; between 0 and 2 of 12 dimensions per seed, most often the happiness emotion dimension) and regenerating the random projection at the same reduced dimensionality: the profile reaches 0.544 ± 0.018 against a matched random projection of 0.607 ± 0.021, a delta of -0.063 ± 0.011, negative in all 5 seeds (one-sided sign test p=0.031). A raw 768-dimensional RoBERTa text embedding — a different, unprojected representation, reported separately rather than as a fourth point on this ladder — reaches 0.584 under the identical fitting protocol, deterministically identical across all 5 seeds.

Because all 5 seeds share the same 47 test participants, a cross-seed comparison alone would propagate only training-randomness uncertainty, not test-subject sampling uncertainty — the source that actually determines whether this generalizes. A subject-level bootstrap (2000 resamples of the 47 test participants, delta between the degeneracy-controlled profile and the matched random projection averaged across all 5 seeds' fixed predictions within each resample) gives mean delta -0.076, 95% CI $[-0.119, -0.037]$, entirely below zero (99.95% of resamples negative). This is the properly-powered statistic for the claim: propagating both training and test-sampling uncertainty, the inversion holds.

Table 1: DAIC test AUROC, mean ± std over 5 seeds, projections of the identical fused representation and fitting protocol. Top: full 12 dimensions. Middle: matched dimensionality after dropping per-seed near-constant profile dimensions (PCA reported at full dimensionality only, having no analogous issue). Bottom: matched-dimension delta, as a cross-seed statistic and a subject-level bootstrap (2000 resamples) propagating training and test-sampling uncertainty.

Representation	DAIC AUROC
Construct profile (full 12-dim)	0.524 ± 0.039
Random projection (full 12-dim)	0.616 ± 0.017
PCA-12 (full 12-dim)	0.610 ± 0.040
Construct profile (matched dim.)	0.544 ± 0.018
Random projection (matched dim.)	0.607 ± 0.021
Δ, matched dim. (cross-seed)	-0.063 ± 0.011
Δ, subject-bootstrap mean (95% CI)	-0.076 $[-0.119, -0.037]$

We checked two alternative explanations before treating this as a construct-supervision effect. First, that it is merely dimensionality reduction of any kind: PCA-12, which is unsupervised, lands within noise of the random projection (0.610 vs. 0.616) and clearly above the construct profile — the effect is specific to construct alignment, not to projecting to 12 dimensions per se. Second, that it reflects zero-shot cross-corpus transfer (the sentiment/emotion heads are trained on MOSEI, not DAIC): we re-derived sentiment and emotion supervision directly on DAIC transcripts with off-the-shelf classifiers (Loureiro et al. 2022; Hartmann 2022) and refit a DAIC-native projector, per seed. Only 1–3 of 7 re-derived dimensions clear the held-out generalization gate per seed — itself informative, since most affective dimensions are not reliably estimable from DAIC's n=107 training set even in-domain. Using whichever dimensions pass, the delta between the in-domain profile and a matched-dimensionality random projection is -0.037 ± 0.084 across the 5 seeds — 3 of 5 negative, 2 of 5 positive (one-sided sign test p=0.5). This control is inconclusive, not corroborating, and does not settle whether domain shift contributes at this sample size; the evidence against domain shift as the primary explanation rests on the cross-corpus finding's own 5-seed replication and subject-level bootstrap, not on this control.

Per-seed DeLong tests comparing the profile against individual random-projection draws reach significance in only 10 of 100 comparisons (5 seeds × 20 draws) — a single such comparison is exactly as underpowered as the absolute decoding claim above, and we report this explicitly as a power limitation rather than as evidence either way; the cross-seed, subject-bootstrapped statistic above is the one we rely on.

One cross-table comparison is worth stating plainly:

Table 2: Graph routing vs. non-graph MMoEEx baseline, mean ± std over 5 seeds (17, 42, 1337, 2024, 31415). Bolded cells reflect the differences ($|\Delta| > \max(\text{baseline CI half-width}, 2 \times \text{pooled seed std})$); all others are within seed noise.

Variant	DAIC AUROC	MOSEI Sent. CCC	FI CCC
Non-graph MMoEEx	0.541 ± 0.017	0.482 ± 0.011	0.571 ± 0.006
V0 (inductive-k10)	0.518 ± 0.027	0.510 ± 0.024	0.480 ± 0.060
V1 (split-local-k10)	0.543 ± 0.033	0.498 ± 0.023	0.481 ± 0.052
V2 (transductive-k10)	0.595 ± 0.063	0.486 ± 0.019	0.493 ± 0.050
V3 (inductive-k15)	0.549 ± 0.057	0.501 ± 0.022	0.507 ± 0.047
V4 (split-local-k15)	0.538 ± 0.051	0.501 ± 0.017	0.506 ± 0.046

the model's own depression head (non-graph baseline, Table 2) reaches DAIC AUROC 0.541 ± 0.017, while an unsupervised random projection of that same model's fused representation reaches 0.616 ± 0.017 (Table 1, full 12-dim) — a supervised depression head underperforming a random projection of its own representation. This is specific to the training regime (the head is optimized under a shared multitask objective, Section 3; the random-projection classifier is fit for depression alone), not a claim that supervision cannot help, but it is the sharpest single statement of this section's result.

5.2 An Independent Dissociation: MPDD

If affective constructs were simply uninformative about depression, the DAIC result above would be unsurprising. MPDD (Fu et al. 2025) lets us test this directly, since it provides real (not model-estimated) Big-Five scores. Ground-truth Big-Five decodes MPDD depression at AUROC 0.667 (95% CI $[0.273, 1.000]$, Young track, n=13 test subjects) and 0.925 (95% CI $[0.727, 1.000]$, Elderly, n=13); pooling both tracks with a stratified estimator (concordant pairs counted only within a track, removing the contribution of between-track differences to the ranking, Section 3.5) gives 0.857 (95% CI $[0.681, 1.000]$, n=26). These intervals are wide and the ceiling-touching upper bound reflects real imprecision at this sample size; we read this arm as corroborating the clinical literature linking personality to depression (Kotov et al. 2010), not as independently establishing it.

The same construct, estimated from MPDD's own raw audio/video features by a ridge regressor trained the ordinary way, fails to generalize at all: held-out R^2 is negative for all five traits (worst: Extraversion, $R^2 = -2.86$; least negative: Agreeableness, $R^2 = -1.75$), despite train-set R^2 of 0.91–0.99 — a textbook overfitting signature at this feature-to-sample ratio (1221 features, 184 training subjects). Ground-truth traits are informative; our estimate of them, in this corpus and at this sample size, is not. We take this as the mechanism behind Section 5.1: the bottleneck is construct measurement, not construct validity.

5.3 Graph-Routing Ablation

The non-graph baseline reaches DAIC AUROC 0.541 ± 0.017 over 5 seeds — a non-chance, working classifier. Applying a reportability rule (a delta counts only if it exceeds both the baseline's own confidence interval half-width and twice the pooled seed standard deviation) to all 20 (variant × metric) cells across DAIC AUROC, MOSEI sentiment CCC, MOSEI emotion AUROC, and FI CCC: on the 15 cells spanning DAIC and MOSEI, none are reportable — every apparent difference from baseline is within seed noise at n=5 seeds; V1 and V3 show a small, non-reportable positive DAIC delta. FI personality is the exception: of its 5 cells, 3 of 5 variants (V0, V1, V2) show a reportable negative delta ($p < 0.05$, paired t-test against the 5 matched seeds), and the remaining two (V3, V4) are borderline ($p \approx 0.05$–0.06) — graph routing consistently and substantially degrades FI personality prediction, the one directional routing effect in this study. A KNN-voting baseline (label smoothing over the same graph neighbourhoods, no learned router), extended to the same 5 seeds, reaches DAIC AUROC 0.623 ± 0.071 — numerically above the non-graph baseline, but not reportably so (paired Δ=+0.083, p=0.087). It is, however, reportably worse than the non-graph baseline on all three other metrics (MOSEI sentiment CCC, MOSEI emotion AUROC, FI CCC; all $p < 0.02$; supplementary material) — simple neighbourhood label-smoothing is not a free improvement once measured across seeds.

A homophily diagnostic (same-dataset edge label agreement/correlation vs. random pairing) clarifies part of this picture, with an important caveat about what representation it measures. The graph actually used to route experts in every V0–V4 run is built over a KNN graph on a separate, randomly-initialized, untrained fusion projection of the raw modality features, not the trained model's own fused representation h_i^{fused} used everywhere else in this paper — a property of the existing graph-construction code that materially changes what the diagnostic can be said to explain, disclosed here rather than carried forward silently. We also fix an unseeded random-projection baseline in this diagnostic, now averaged over the 5 canonical seeds. Table 3 reports both the untrained-embedding version and a version recomputed on the trained representation. On the untrained embedding actually used for routing, the 5-seed result is unambiguous: DAIC shows no signal beyond chance in any of the four leakage-safe variants ($p \in [0.12, 0.78]$), while MOSEI and FI both show strong signal in every variant ($p < 0.0001$). Recomputing on the trained representation instead (6 checkpoints × 4 variants) shows a more specific pattern: DAIC signal is graph-topology-dependent (inductive variants V0/V3, Δ=+0.064/+0.042, both $p < 0.0001$; split-local V1/V4 show no separation, p=0.36/0.94), while MOSEI and FI again show strong signal regardless of topology.

Two targeted fixes were tried against the (untrained-embedding) DAIC null, both extended to the full 5-seed protocol (mean ± std; same reportability rule

Table 3: Real graph neighbours vs. random pairing. "Untrained embedding" is the representation used to build every V0–V4 routing graph (random projection of raw features, averaged over the 4 leakage-safe variants). "Trained representation" recomputes the same diagnostic on the corresponding checkpoint's fused representation (mean over 6 checkpoints × 4 variants; DAIC split by graph topology, Section 5).

Task (representation)	Real graph	Random
DAIC, untrained embedding (5-seed mean)	0.564	0.561
DAIC, trained repr. (inductive: V0/V3)	0.619	0.566
DAIC, trained repr. (split-local: V1/V4)	0.555	0.562
MOSEI, untrained embedding (5-seed mean)	0.249	−0.001
MOSEI, trained repr. (correlation)	0.485	0.000
FI, untrained embedding (5-seed mean)	0.289	−0.000
FI, trained repr. (correlation)	0.423	−0.001

and same non-graph baseline as Table 2). A $K \in \{5, 10, 15, 20\}$ sweep (inductive graph; K=10 and K=15 are V0 and V3 above) shows no monotonic trend and no configuration's DAIC delta from baseline is reportable (K=5: 0.541 ± 0.048, Δ=+0.001, p=0.97; K=20: 0.526 ± 0.057, Δ=−0.015, p=0.55); K=5 does show a reportable negative FI delta (p=0.04), consistent with the FI-degradation pattern above, while K=20's FI delta is borderline (p=0.06). A learned per-task routing weight (replacing the fixed λ=0.5) reaches DAIC AUROC 0.546 ± 0.034; the delta from the fixed-λ V0 configuration is +0.028, not reportable under this effect-size criterion despite reaching p=0.007 on a paired t-test (supplementary material). Neither fix recovers a reportable DAIC benefit, consistent with the diagnosis that the routing graph itself lacks the relevant signal.

6 Discussion

The construct-bottleneck and graph-routing results are independent findings sharing a protocol and a dataset, not a causal chain: the V0–V4 routing graph is built on an untrained projection, not the trained representation h_i^{fused} that Section 5.1 analyzes, so that result cannot explain why the untrained graph fails — a random projection is not construct-organized, and Section 5.1 shows random projections preserve depression-relevant variance better than a construct-aligned one, if anything making the untrained graph's failure more puzzling. Worth stating plainly: decodability (can a classifier recover depression from a representation) and local homophily (do a sample's neighbors share its label) are different properties — the trained representation's topology-dependent DAIC signal (Section 5, Table 3) coexists with that same representation's construct-projection failing to decode depression above chance. FI makes the point most directly: both homophily measures find strong real signal for FI in every graph variant, yet routing reportably degrades FI prediction in 3 of 5 variants ($p < 0.05$, other two borderline) — not signal absence, but a supervised transformation converting available structure into a worse prediction than not using it at all, the one directional routing effect in this study and the clearest single instance here of supervised structure being actively harmful rather than merely unhelpful. We read the routing and construct-bottleneck results as instances of a common theme — supervised structure fit at small clinical sample sizes can destroy more than it organizes — rather than one confirming the other's mechanism. The leakage-safe protocol remains essential regardless: transductive (non-leakage-safe) construction reaches the highest DAIC AUROC of any variant (0.595 ± 0.063, Table 2); this delta is not reportable, and we draw no inference from it beyond noting the direction.

MPDD and DAIC fail for different reasons. MPDD is classical attenuation: self-report measurement error shrinks the observed correlation between an estimated construct profile and the true trait, a channel-noise effect documented since Spearman (1904). DAIC's failure is not attenuation: the construct and random projections act on the identical, error-free representation, so no noisy channel intervenes. Instead the construct projection is an information bottleneck, fit to preserve variance aligned with its own supervised targets and discarding the rest, while a random projection preserves those discarded directions incidentally, retaining more depression-relevant variance than the projection built to predict depression. Both are consistent with the bottleneck being measurement, not validity, but the mechanisms differ: attenuation predicts recovery from better items, an information bottleneck only from changing what the projection is trained to preserve.

Limitations: DAIC's small train (n=107)/test (n=47) sets limit power for absolute claims, unlike the better-powered comparisons carrying this paper's evidence (Section 5.1); reliance on frozen pretrained encoders; graph-quality sensitivity when modalities are missing; and untried remedies beyond our two fix attempts (jointly-trained embeddings, graph rewiring, task-affinity grouping; supplementary material). DAIC runs text-only under native routing (Section 3), so the multimodal architecture does no work for either headline DAIC result. The in-domain control is inconclusive, not corroborating; the cross-corpus and MPDD results, both 5-seed replicated, carry the paper's weight. LLM encoders, calibration, domain adaptation, and XAI explored during development touch none of these results and are reported separately as single-seed and exploratory (supplementary material).

7 Conclusion

This paper reports two independent findings on a common protocol and dataset. First: graph-guided MoE routing does not improve depression detection across five leakage-safe configurations, a K-sweep, and a learned routing weight; a homophily diagnostic localizes why on the untrained embedding actually used for routing, with a more specific topology-dependent pattern on the trained representation, and FI personality shows the one directional routing effect — a case of supervised structure actively destroying signal shown present by the same diagnostic. Second, examined separately, affective-construct projections (sentiment, emotion, perceived personality) achievable at clinical sample sizes degrade depression-relevant variance that an unsupervised projection of the same shared representation retains — a matched-dimensionality comparison, replicated across 5 seeds, resilient to dropping near-constant dimensions, negative under a subject-level bootstrap. A companion in-domain re-derivation is itself inconclusive under proper seed replication (3 of 5 seeds negative, 2 of 5 positive) — so this claim is about achievable construct estimation, not construct supervision failing in principle. An independent corpus (MPDD) localizes a convergent failure to construct estimation for a different reason (data starvation rather than cross-corpus transfer): ground-truth Big-Five traits decode depression, while traits estimated from the same corpus's own features do not generalize at all. Two different mechanisms degrading construct estimation is what carries the general claim: the bottleneck is construct measurement, not construct validity, consistent with the psychological literature on affect and depression. We recommend validating any intermediate predictor on held-out data before consuming its output downstream, preferring variance-preserving bottlenecks over construct-supervised ones at small clinical sample sizes, and reporting an unsupervised-projection control alongside any supervised-representation claim. We release the leakage-safe graph protocol and the full ablation ladder.

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